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Supplementary Material
Viewpoint Invariance Is a Theorem, Not a Fit

Anonymous WACV Applications Track submission

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This supplement carries the depth condensed out of the main paper: the proof of the invariance theorem (Sec. S1), the full invariant-set specification (Sec. S2), the eight-gate numerical certificate (Sec. S3), the complete precision budget (Sec. S4), the nondeterminism seed distribution (Sec. S5), the protocol audit and continuous-time control (Sec. S6), and the units audit behind our comparison against published KIMORE numbers (Sec. S7). All numbers are reproduced from the same banked artifacts as the main paper; equation and proposition references (e.g. Prop. 1) point to the main text.

S1. Proof of the Invariance Theorem

Proposition 1 ($SO(3)$-invariance of the read-out; = Prop. 1)**.** *Let the encoder intertwine with the Wigner-D action, so under $g \in SO(3)$ each latent transforms as $s_j \mapsto s_j$ (type-0) and $v_j \mapsto D_1(g)\, v_j$ (type-1), with $D_1(g) \in SO(3)$ orthogonal and $\det D_1(g) = +1$. Then the projection Π (main Eq. 1–2) is invariant under $SO(3)$ for arbitrary encoder weights, and the predicted score is invariant to any rigid camera motion.*

Proof. Feeding root-relative coordinates makes every input a difference $\tilde{\boldsymbol{x}}_i - \tilde{\boldsymbol{x}}_j$, so a camera translation cancels; it remains to treat rotations. Consider each generator of Eq. 1 under $v \mapsto D_1(g)v$ with $R := D_1(g)$ orthogonal: (i) $\tanh s_j$ is a function of a type-0 scalar, hence unchanged. (ii) $\|v_j^c\| = \|Rv_j^c\|$ since $R^\top R = I$; so $\log(1 + \|v\|)$ is invariant. (iii) $\langle \widehat{Rv}, \widehat{Rv'} \rangle = \langle Rv/\|v\|, Rv'/\|v'\| \rangle = \langle \hat{v}, \hat{v}' \rangle$, again because R preserves inner products. (iv) $\det[\widehat{Rv^a}, \widehat{Rv^b}, \widehat{Rv^c}] = \det(R)\det[\hat{v}^a, \hat{v}^b, \hat{v}^c] = \det[\hat{v}^a, \hat{v}^b, \hat{v}^c]$, using $\det(R) = +1$; the triple products are invariant under *proper* rotations (and flip sign under a reflection, $\det = -1$, which is exactly why they narrow the guaranteed group from $O(3)$ to $SO(3)$). The bone lengths $d_{ij} = \|\tilde{\boldsymbol{x}}_i - \tilde{\boldsymbol{x}}_j\|$ and the anatomical signed volumes χ_τ (normalised determinants of joint triads) are invariant by the same orthogonality and $\det(R) = +1$ arguments. Finally mean_j and $\max_j$ are applied identically to invariant per-joint quantities, hence themselves invariant. Every entry of Π is therefore fixed by g, for any weights; the head is a deterministic function of Π, so the score is invariant. □ □

Table S1. Structural specification of the invariant cut Π.

Generator	Irrep	Count	Dim
Parity-even — generating set complete for $O(3)$			
scalars $\tanh s$	$0e$	n_0	32
vector norms $\log(1{+}\|v\|)$	$0e$	n_1	8
cosines $\langle \hat{v}, \hat{v}' \rangle$	$0e$	$\binom{n_1}{2}$	28
pooled mean $\oplus$ max		2×68	136
bone lengths d_{ij}	$0e$	per bone	24
even subtotal			**160**
Parity-odd — $SO(3)$ chiral completion			
triple products $\det[\hat{v}, \hat{v}', \hat{v}'']$	$0o$	$\binom{n_1}{3}$	56
pooled mean $\oplus$ max		2×56	112
anatomical volumes χ_τ	$0o$	per frame	11
odd subtotal			**123**
full $SO(3)$ cut			**283**

S2. Full Invariant-Set Specification

Table S1 states both generating sets explicitly at the deployed widths ($n_0{=}32$, $n_1{=}8$; $J{=}25$ joints, 24 bones, $|\mathcal{T}|{=}11$ anatomical frames). Both parity classes share one $\mathrm{mean} \oplus \max$ joint pool; bone lengths and signed volumes are concatenated index-preserved (unpooled). The parity-even subtotal (160) is exactly the projection whose precision budget is audited in Sec. S4; the parity-odd completion adds 123 dimensions.

S3. Numerical Certificate (E1–E8)

We certify the *deployed* model. E1–E5 establish that the claimed invariance is present; E6–E8 establish that it is the one we actually have. The three conditions of E3–E4 — violation at roundoff, *flat* in angle, and $r \gg 1$ — are what upgrade "small" to "zero": magnitude alone would not. Reported figures are from a representative fp64 run; being roundoff-floor quantities they are *regime*-stable (order of magnitude, sign of slope, $r \gg 10^5$) but not bit-reproducible.

E1 **Representation orthogonality:** $\max_g \|\rho(g)^\top \rho(g) - I\|_F \leq 10^{-12}$ (fp64).

E2 **Invariance of the motion channels:** δ, σ verified invariant, not assumed.

E3 **Rotation sweep:** score violation $\sim 10^{-16}$ across $\theta \in [0.05, 3.0]$ rad, with log–log slope $|\text{slope}| < 0.5$ (flat). A modelling violation grows with angle; roundoff does not.

E4 **Precision scaling:** $r = \bar{\delta}^{\text{fp32}}/\bar{\delta}^{\text{fp64}} = 5.44 \times 10^{-8}/4.43 \times 10^{-16} = 1.23 \times 10^8$, tracking the machine-epsilon ratio; a genuine violation would be precision-independent ($r \approx 1$).

E5 **Translation invariance:** on raw world coordinates, $|s(\boldsymbol{x}+\boldsymbol{c}) - s(\boldsymbol{x})|/|s(\boldsymbol{x})| = 2.6 \times 10^{-16}$.

E6 **Parity:** under a reflection the parity-even cut gives *exactly* 0 (bit-identical on a mirrored body — an unintended $O(3)$-invariance), whereas the completed cut gives 4.4×10^{-2} (seed 0, `certify_egru.json`). A single reflection certifies the entire improper coset.

E7 **Laterality:** permuting left/right joint *indices* (a relabelling, not a reflection) moves the score by 6.9×10^{-2} / 5.6×10^{-2} for the two cuts (seed 0); both separate a left- from a right-limb movement. E6–E7 are model-dependent modelling quantities, not roundoff-floor ones, and vary across seeds.

E8 **The fix is free:** read-out invariance under Haar-random *proper* rotations (up to $\theta{=}\pi$), with the parity-odd channels active, is 2.9×10^{-16} — unchanged. E1–E5 are likewise over Haar-random $SO(3)$, so the certified invariance is to the full rotation group, of which a camera azimuth or the NTU cross-view displacement is one instance. We report no separate NTU-specific invariance measurement: the gates are architectural and the corpus does not change them.

S4. Complete Precision Budget

The full fp64→int8 budget — including the bf16 and int8-activations-only rows — is reported in the main paper's deployment table. The int8 cliff is attributable *solely* to activation grid-rounding: weight-only quantization preserves the theorem (a quantized model is a different equivariant model), and fp16 is 3.2× tighter than bf16 because equivariance pays in mantissa bits, which bf16 trades for exponent range.

S5. Nondeterminism: Seed Distribution

The 0.33 MAD floor quoted in the main paper is the tighter *fixed-configuration* run-to-run component (init held constant, atomic nondeterminism only). Table S2 gives the wider *seed-to-seed* distribution (SO(3) EGRU clean MAD per fold across three initializations): mean across-seed std 0.48, worst pairwise gap 1.30. Both exceed every model-vs-model clean gap in the main accuracy table, so those gaps are ties. cuDNN-deterministic modes remove the cuDNN-GRU atomic component but not the initialization variance; the structural degradation columns are weight-independent (Prop. 1 holds for arbitrary weights), so the flat viewpoint curve stayed exactly flat across every seed while absolute MAD moved.

Table S2. Per-(seed,fold) clean MAD, SO(3) EGRU. No retraining beyond the three seeds.

fold	seed 0	seed 1	seed 2	std	gap
0	7.120	6.778	6.737	0.210	0.383
1	6.680	6.124	5.804	0.443	0.876
2	6.976	7.345	6.623	0.361	0.722
3	6.762	7.363	6.093	0.635	1.270
4	6.469	7.714	6.415	0.735	1.299
mean across-seed std				**0.48**	

S6. Protocol Audit and Continuous-Time Control

Protocol audit. The single-exercise audit — selection bias $+1.2 \pm 0.4$ MAD (18.6% over three seeds), paired subject-cluster bootstrap $\Delta = -0.44$ (per-seed $[-1.07, +0.20]$, clearing zero in only one of three seeds) — is reported per-seed in the main paper. It is *not* subject leakage (each subject contributes one recording on that slice, so a random split is already subject-disjoint); the slice simply carries no robust subject-level signal, so we pool the five exercises and gate every experiment on a per-exercise mean-predictor floor.

Continuous-time control. We implement the Neural CDE variant and certify it to the same E1–E8 standard; it satisfies an exact equivariance certificate and *fails* its own measured floor (8.43 vs. 8.25 MAD), the cleanest demonstration that a structural guarantee is worthless without measured signal. Analysing the adaptive solver, a Dormand–Prince controller accepts/rejects steps on a per-component scaled-RMS norm that is *not* group-invariant even when the vector field is exactly equivariant, so rotated and unrotated integrations receive different step grids and de-align; a per-irrep-normalised norm removes this, driving grid divergence to identically zero.

Irregular-sampling null (derived). A model-free spectral census finds 97.7% of KIMORE's *positional* energy below the fixed-grid resampling corner (though 70% of *velocity* energy lies above it), so resampling preserves pose

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$\widehat{\lvert y \rvert} = 100\,\mathrm{MAD/MAPE}$	Ex1	Ex2	Ex3	Ex4	Ex5
Du et al. 2021	39.4	36.6	32.8	34.1	33.2
Deb et al. 2022	41.5	60.8	51.4	42.1	39.0
Yao et al. 2023	40.2	35.1	32.5	35.5	36.1
Mourchid et al. 2023	39.5	77.3	34.3	38.1	32.8
Zhang et al. 2025	41.2	51.6	38.4	42.2	35.0
Kuang et al. 2025	43.2	31.4	41.0	30.6	32.2
Kuang et al. 2026	54.5	40.1	48.3	63.4	18.8
Point-cloud transf.	34.1	29.6	38.1	33.4	32.4
true score mean	**40.7**	**35.6**	**39.5**	**35.0**	**36.4**
true $\mathrm{mean}\lvert z \rvert$	0.80	0.87	0.79	0.86	0.84

Table S3. Target magnitude recovered from each paper's own published MAD/MAPE pair, against KIMORE's true clinical Total Score statistics. Every row lands in the score band (30–63), none anywhere near the standardised band (≈ 0.8). Rows reporting no MAPE are omitted.

geometry while low-passing the derivative band and acts as a denoiser on healthy-form exercises, where the discriminative signal is positional — the null was predictable before the experiment, though the boundary would move for tremor-dominated tasks. The mechanism is nonetheless real: a physiologically calibrated 5 Hz tremor is recovered from the true irregular stamps at $r{=}1.000$ under drop rates up to 70%, while a resampled estimator decays to 0.785.

S7. Units of the Published KIMORE Numbers

The main text compares our MAD against published KIMORE MAD directly. That is only legitimate if both are in clinical-score units, and none of those papers states the scale. The question is not academic: the reference point-cloud transformer standardises its targets with a `StandardScaler` ($\sigma = 8.466$ on `Kimore_ex1`) and its per-epoch training print is in standardised units while its evaluator inverse-transforms, so the two readings of the same headline number differ by $8.5\times$.

The published tables settle it without recourse to anyone's source code. Every method in this lineage reports MAD and MAPE for the same run, with $\mathrm{MAPE} = 100 \cdot \mathrm{mean}(\lvert y - \hat{y} \rvert / \lvert y \rvert)$, so

$$\widehat{\lvert y \rvert} \;=\; 100 \cdot \mathrm{MAD/MAPE} \tag{1}$$

recovers the magnitude of whatever array was handed to the metric. On raw clinical scores it must land near KIMORE's per-exercise means; on standardised targets it must land near $\mathrm{mean}\lvert z \rvert \approx 0.8$.

Table S3 is unambiguous: the median recovered magnitude over all 40 published cells is 38.1 against a true per-exercise mean of 35.0–40.7 (median relative error 10.6%), while the standardised reading is excluded by at least $70\times$ — it would require these papers to have printed MAPE values in the tens of percent rather than the 0.17–6.0 they report. The published KIMORE MAD numbers are therefore in 0–50 clinical-score units, and our 6.3–6.7 is on the same axis. The audit is reproduced by `src/published_units_audit.py`, which gates on all three conditions.